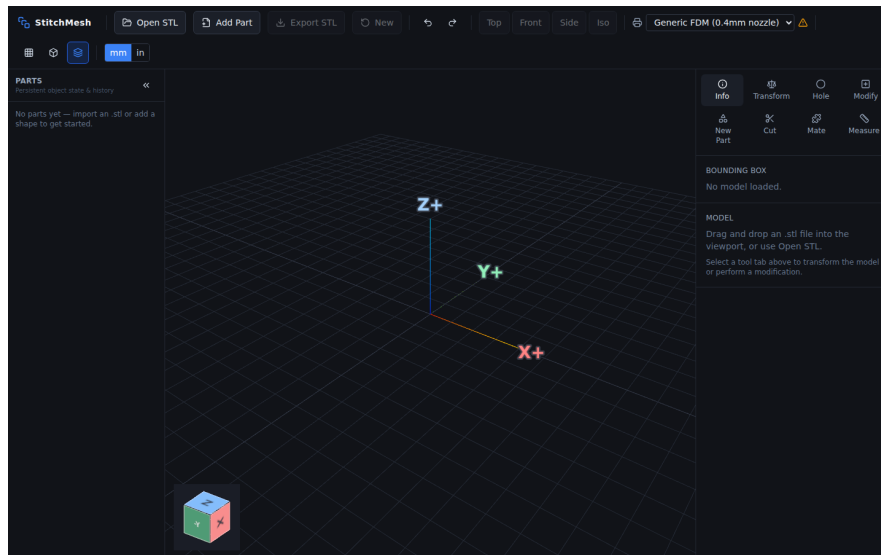




StitchMesh

Complete User Manual

“Modify, don’t model.”

An offline tool for making localized edits to an existing .stl file — resize a hole, add a threaded boss, cut a model in two, measure a clearance — without learning a full CAD package.

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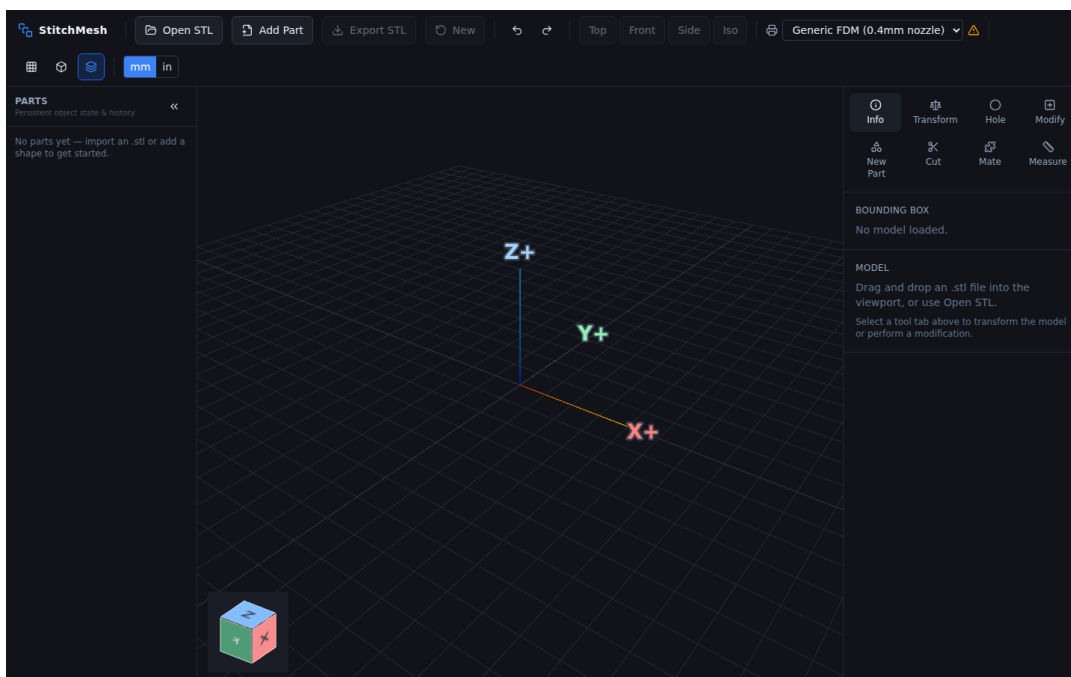
> **Release update: Parts-first workflow** — this edition documents the persistent Parts command center, visibility controls, part locking, context actions, feature visibility/naming, and the combined Transform/Move workspace.

"Modify, don't model." StitchMesh is an offline tool for making localized edits to an existing `.stl` file — resize a hole, add a threaded boss, mate and weld two parts together, cut a model in two and move the pieces apart, measure a clearance — without learning a full CAD package.

This manual walks through every feature with screenshots so you can find what you need quickly.

1. What StitchMesh is

A four-panel app: a **toolbar** across the top (file, undo/redo, view, units, printer, display toggles), a collapsible **Parts panel** on the left listing every object in the scene as its own layer, a **3D viewport** filling the center, and a **context sidebar** on the right with nine tool tabs whose contents change based on which is active: **Info**, **Transform**, **Hole**, **Primitive**, **Shapes**, **Cut**, **Mate**, **Move**, **Measure**.



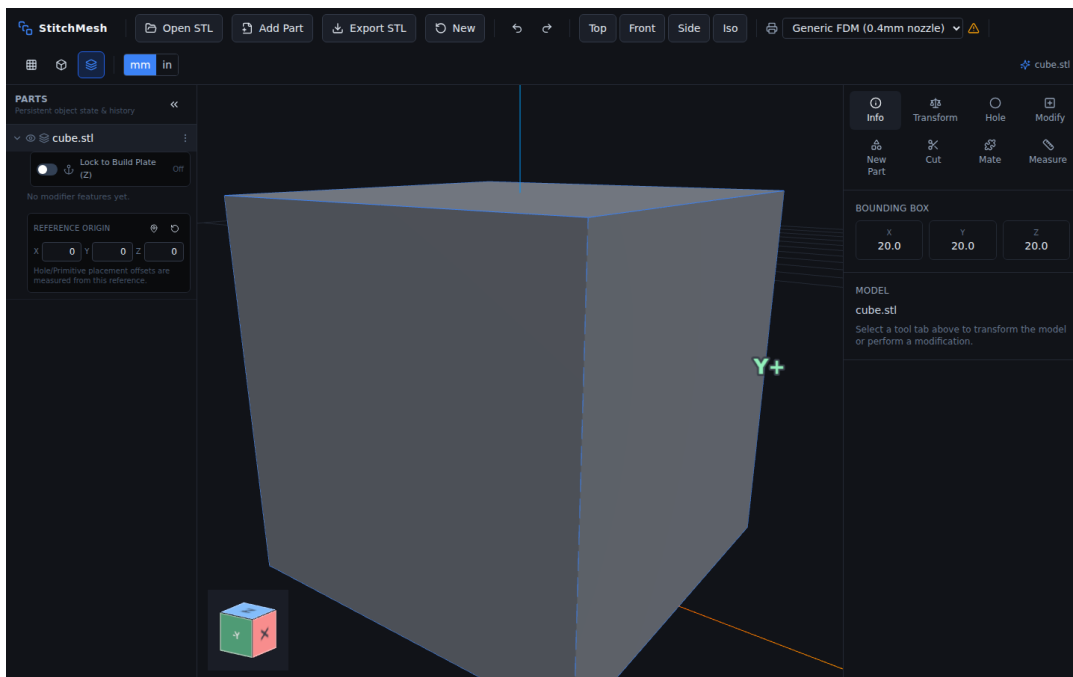
Empty state. Toolbar across the top (Open STL, Add Part, Export STL, New, Undo/Redo, view presets, unit toggle, printer picker, display toggles). Parts panel on the left, empty. Dark viewport with a grid, an axes gizmo, and the orientation cube in the bottom-left corner. Sidebar defaulted to Info, reporting "No model loaded."

StitchMesh is a single self-contained HTML file. Double-click it to open it in your browser and it works fully offline — no install, no server, and nothing is ever sent over the internet. Importing reads your `.stl` file straight into memory, and exporting saves a new file through your browser's normal download flow, so your source file is never modified.

2. Loading, adding, exporting, and resetting a model

- **Import:** drag an `.stl` onto the viewport, or click **Open STL** to pick a file. This *replaces* everything currently in the scene. Reading a file never modifies it on disk.
- **Add Part:** imports another `.stl` as a second, independent object next to the first, rather than replacing it — see §5 (Parts Panel). The **Shapes** tab (§9) does the same thing for basic solids instead of an imported file. If nothing is loaded yet, Add Part behaves exactly like Open STL.
- **New:** clears the current model (and its undo history). Asks for confirmation first if a model is loaded.
- **Export STL:** saves the current scene — every part in it — as a new file, downloaded as `<original-name>-modified.stl`. Your source file is never overwritten.

Printer-not-selected export gate: the first time you click Export STL without having picked a printer profile (§18), a dialog interrupts: *"No printer selected yet... Click OK to export anyway with Generic, or Cancel to pick your printer first."* Accepting confirms Generic so it won't ask again.



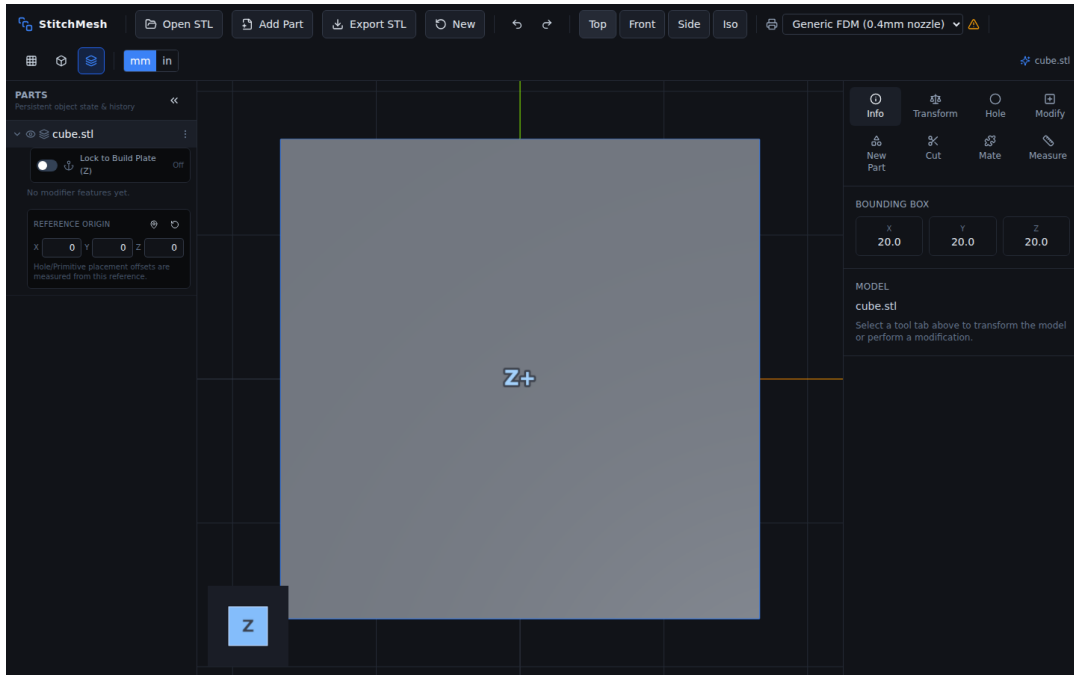
After import: the model renders, the part appears in the Parts panel on the left, and the Bounding Box readout (visible on every tab) reports the **selected part's** live $X \times Y \times Z$ dimensions.

3. Viewport navigation

- **Orbit:** middle-click-drag only. It re-centers on whatever's under the cursor the moment you press the button, rather than orbiting around a fixed point — click near a corner and drag, and the view pivots around that corner.
- **Pan:** right-click-drag.
- **Zoom:** scroll wheel.
- **Left-click** is deliberately *not* an orbit control — it's reserved for tool actions (Hole, Primitive, Mate, Measure, picking a local origin) and for directly dragging a part that's **locked to the build plate** across it (§5, §12).

Left-clicking empty space or an unlocked part does nothing.

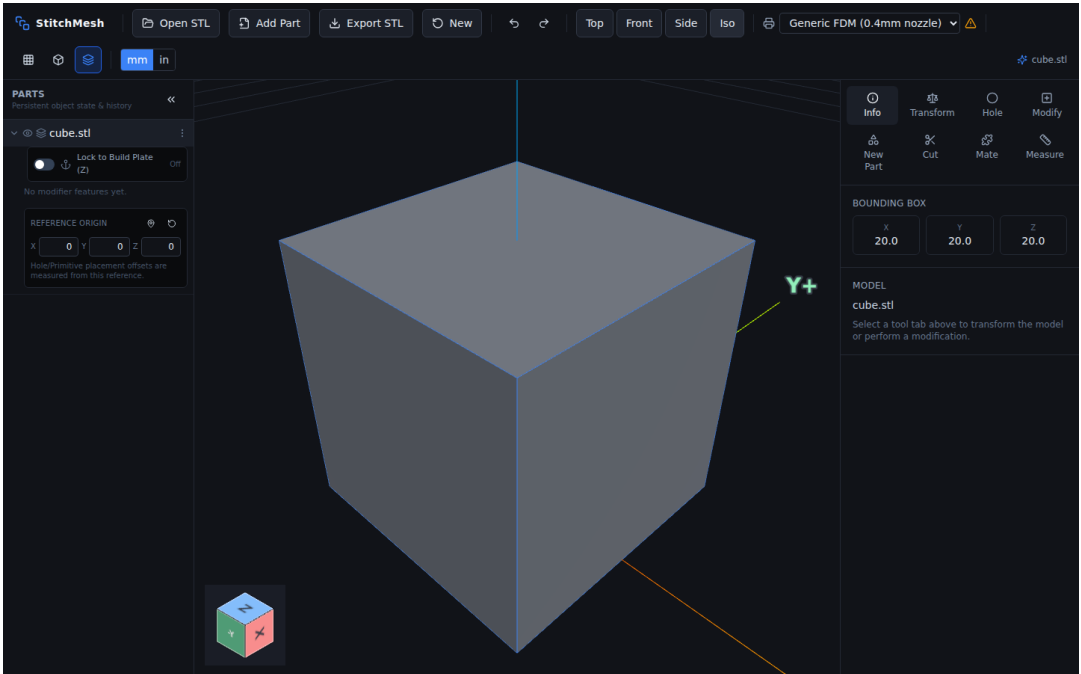
- **View presets** (toolbar): **Top / Front / Side / Iso**, or press **1 / 2 / 3 / 4** on the keyboard.



*Top preset. Presets re-frame and re-target the camera to the whole scene's **current** bounding box, so they stay useful after scaling, moving parts apart, or cutting.*

Orientation cube: the small cube in the viewport's bottom-left corner always shows the current view orientation, rotating in sync with the main view — turn the model and the cube turns with it. Each face is labeled: the bright **X** / **Y** / **Z** faces point along the positive axis, the matching darker **-X** / **-Y** / **-Z** faces point the other way.

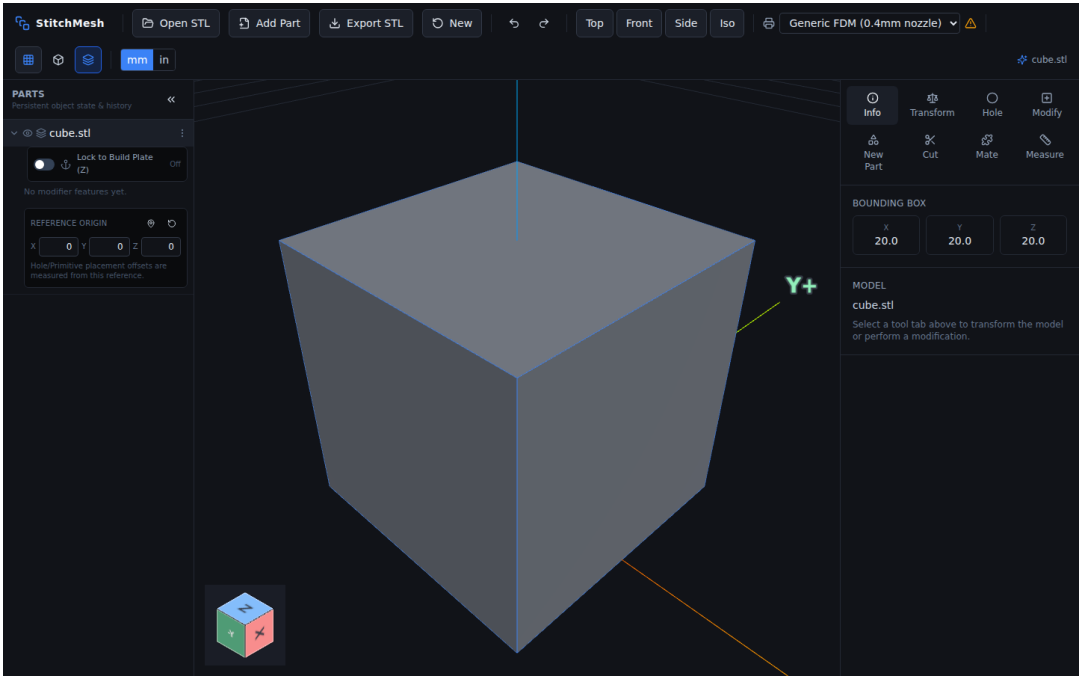
World axis lines: three colored reference lines run from the build plate's origin — red for X, green for Y, blue for Z — and each one is labeled **X+**, **Y+**, or **Z+** right at its tip, so you can read the scene's orientation directly off the lines themselves, not just the corner cube. The label always faces the camera and stays legible even when a part sits in front of it.



Each axis line ends in a colored, always-camera-facing "+" label — X/Y/Z orientation is readable at a glance from anywhere in the scene.

Display toggles (toolbar icons, or keyboard **W** / **B**):

Icon / key	Toggle	Effect
Grid icon / W	Wireframe	Renders the model as an edge mesh
Cube icon	Flat shading	Faceted instead of smooth shading — shows actual triangle density
Layers icon / B	Bounding box	Shows/hides the blue outline around the selected part (on by default)



Wireframe on, from the Iso preset. The orientation cube sits in the bottom-left corner throughout.

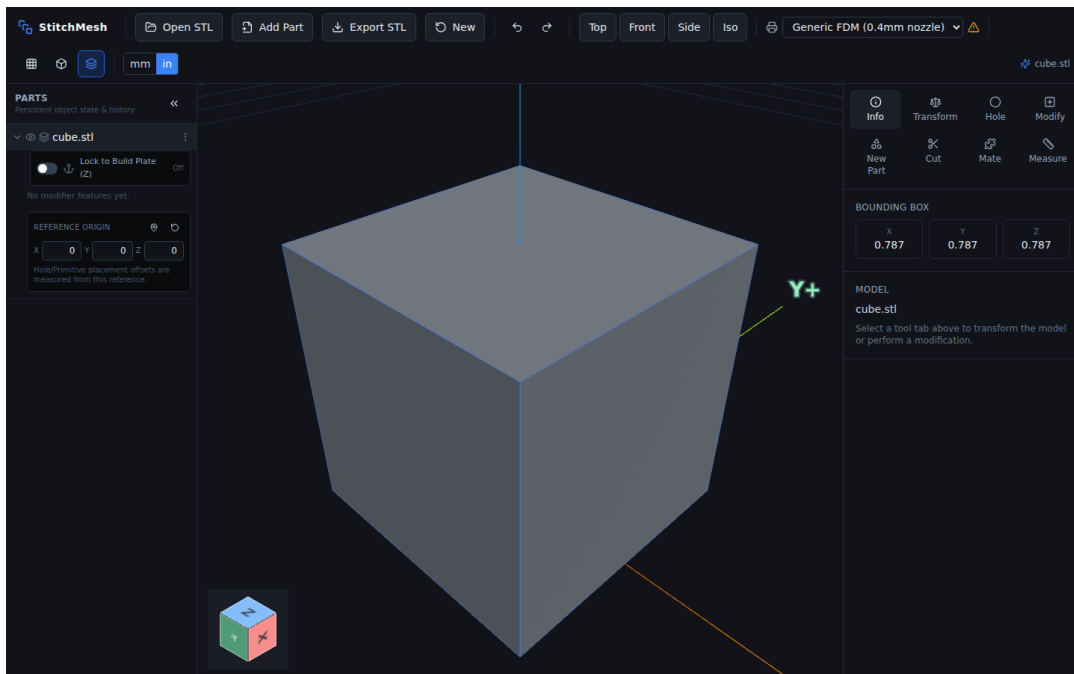
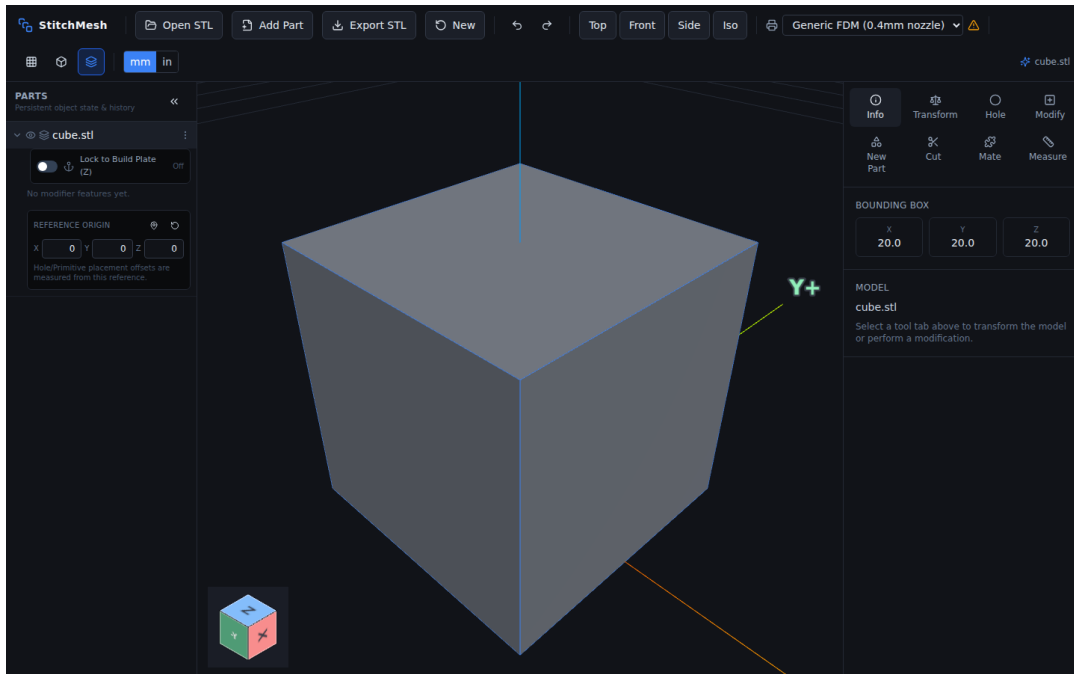
Keyboard shortcuts reference

Key	Action
Ctrl/Cmd + Z	Undo
Ctrl/Cmd + Y, or Ctrl/Cmd + Shift + Z	Redo
1 2 3 4	Top / Front / Side / Iso
W	Toggle wireframe
B	Toggle bounding box

Shortcuts are suppressed while typing in any text field or dropdown.

4. Units — millimeters or inches

The **mm / in** toggle in the toolbar switches every length field in the app — Bounding Box, Scale, Position, Diameter/Depth, Local Origin, Offset-from-origin, and Measure — between millimeters and inches.

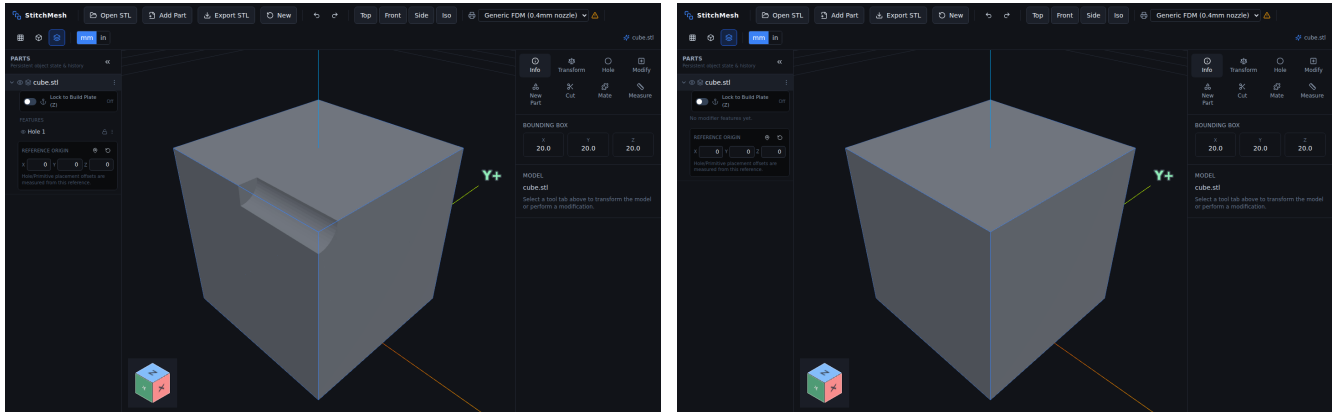


The same 20mm cube: Bounding Box reads 20.0mm in the left shot, 0.787in in the right after switching units. Nothing about the model itself changes — StitchMesh always stores geometry in millimeters internally; the toggle only changes how numbers are displayed and typed.

You can type in either unit at any time — switch to inches, type a diameter in inches, switch back to mm, and the value converts correctly both ways.

5. Undo / Redo

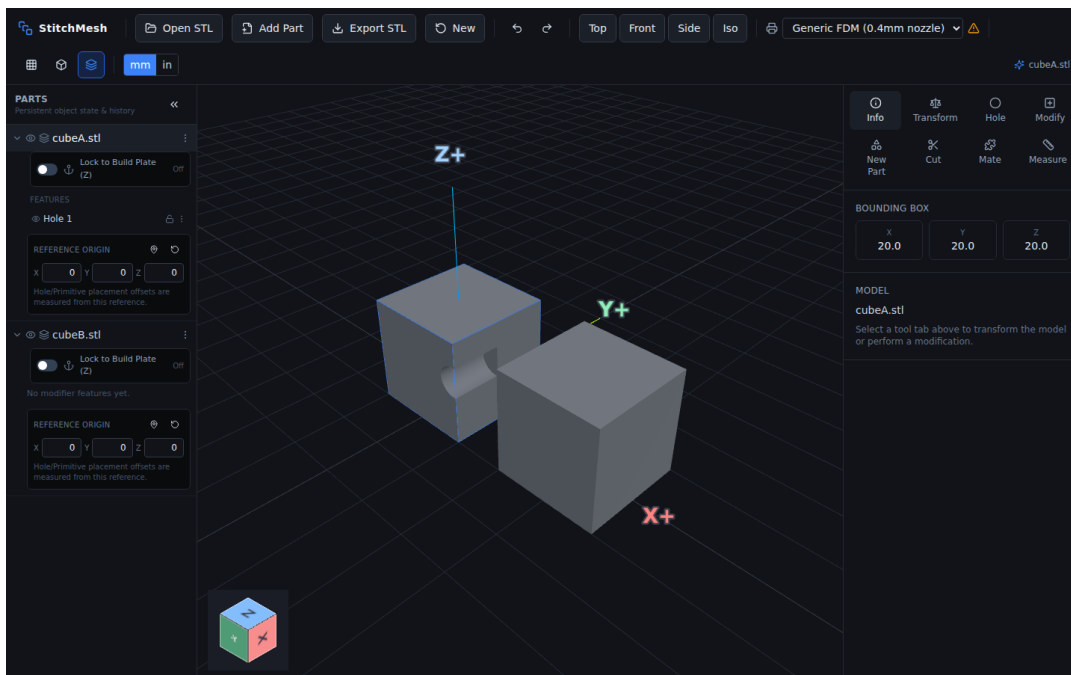
Toolbar: the two curved-arrow icons next to New, or **Ctrl+Z** / **Ctrl+Y**. Every committed change is undoable: Hole/Primitive/Plane Cut/Mate/Weld apply, feature edits and deletes, Transform actions (scale, rotate, center, drop, mirror), Move (including a drag), adding a part or shape, and mesh repair. Continuous edits (typing digits into a field, or a drag) are coalesced into a single undo step per edit session rather than one step per keystroke/frame; discrete actions (a button click) are always their own step.



Left: a hole applied. Right: immediately after **Ctrl+Z** — back to the unmodified cube. Redo (button or **Ctrl+Y**) reapplies it.

6. Parts panel: layers, features, local origin, and lock to plate

The **Parts** panel on the left lists every object currently in the scene — the first import and anything added afterward — as its own **layer**, collapsible via the « button in its header.



Two independent parts. Each part has a dedicated ****Lock to Build Plate (Z)****

toggle directly beneath its name. Feature controls remain nested under the part, and each part also carries its own Reference Origin editor.*

Each part is fully independent — selecting a part (click its row, or use the **Part** dropdown on the Move/Cut tabs) makes it the target for Transform, Move, Mirror, and Plane Cut; none of those actions touch any other part.

Persistent part controls

The Parts panel is now the command center for state that belongs to an object rather than to a temporary tool. Each part row can show:

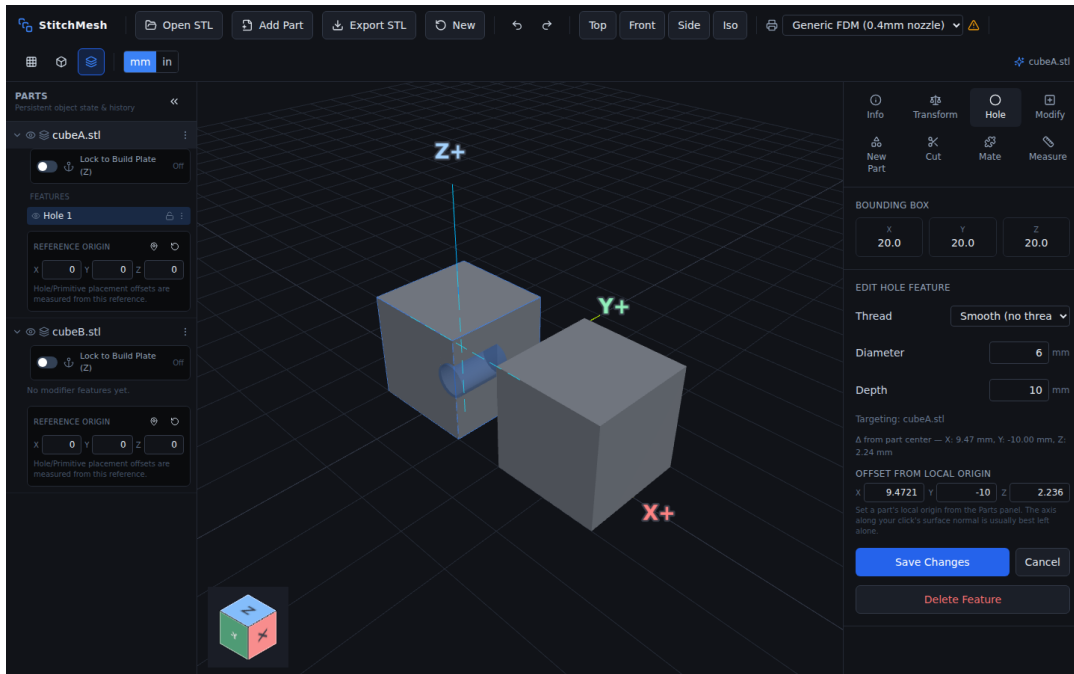
- **Eye** — show/hide the part without deleting it.
- **Lock to Build Plate (Z)** — the per-part toggle directly beneath the part name. This pins the part to Z=0 while keeping X/Y movement available.
- **Part lock** — freezes direct movement and transformation until unlocked. Feature editing remains available, so you can protect placement without making the model unusable.
- **Actions menu** — rename, duplicate, isolate, show all, lock/unlock, lock/unlock to plate, jump directly to Transform/Hole/Modify, or delete the part.

Right-clicking a part opens the same action menu. Double-clicking a feature name lets you rename it.

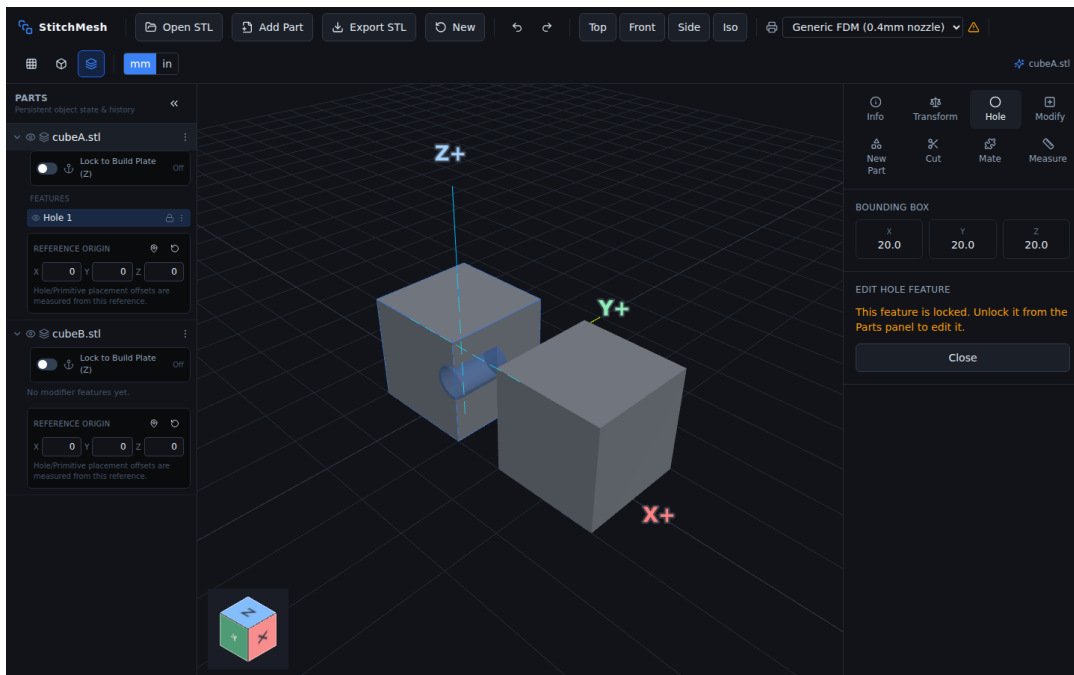
Feature layers

Every Hole and Primitive you apply becomes its own named layer nested under its part ([Hole 1](#), [Add cylinder 1](#), [Cut cylinder 2](#), ...) instead of being baked in and forgotten:

- **Click a feature's name** to reopen it in the Hole/Primitive panel with its exact diameter, depth, thread, and position — change anything and click **Save Changes**, or **Delete Feature** to remove it entirely.
- **Eye icon** — temporarily hides/shows the feature. Hidden features are excluded from the live composite while the base mesh remains visible.
- **Lock icon** — freezes a feature (including its thread/diameter) so it can't be accidentally changed; click the feature and every field is replaced with a locked notice until you unlock it again from the same icon.
- **More (...) menu** — edit/focus, rename, hide/show, lock/unlock, or delete the feature.
- **Delete icon** (trash) — removes that one feature immediately (fully undoable — [Ctrl+Z](#) brings it right back).



Clicking **Hole 1** reopens it for editing: the sidebar shows its current Diameter/Depth/Thread, a Δ -from-center readout, and Offset-from-origin fields. Two dashed cyan lines mark the part's own center on the two axes not aligned with the hole's drilling direction, so you can see at a glance how far off-center it sits.



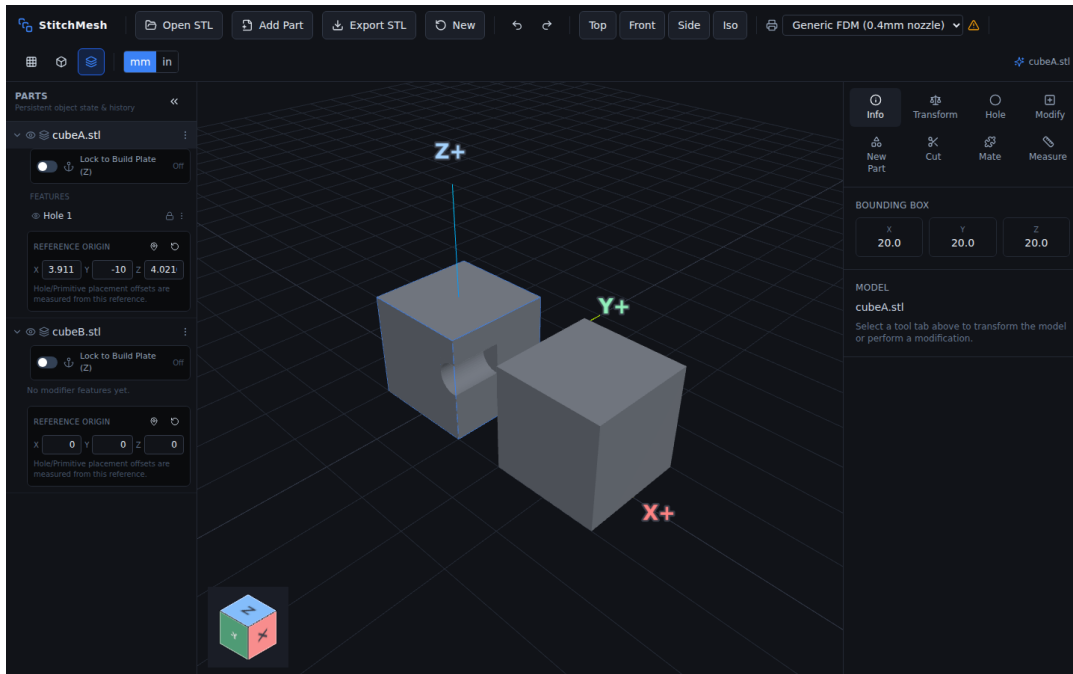
Reopening a locked feature shows a notice instead of editable fields. Unlock it from the Parts panel (the same lock icon) to make changes again.

One tradeoff worth knowing: features stay independently editable only as long as you don't change that part's scale. Scaling a part (Transform → Scale) permanently folds every existing feature on that part into its base shape — the holes/bosses stay exactly where they are, just no longer individually re-editable afterward.

Local origin

Each part can have its own reference point — independent of world (0,0,0) — for typing exact Hole/Primitive positions instead of eyeballing a click. Under a part's feature list:

- **Pin icon** — click, then click a point on that part in the viewport to set the origin there (e.g. a corner of a block).
- **Reset icon** — puts the origin back at the part's own center.
- **X / Y / Z fields** — type an exact origin position directly.



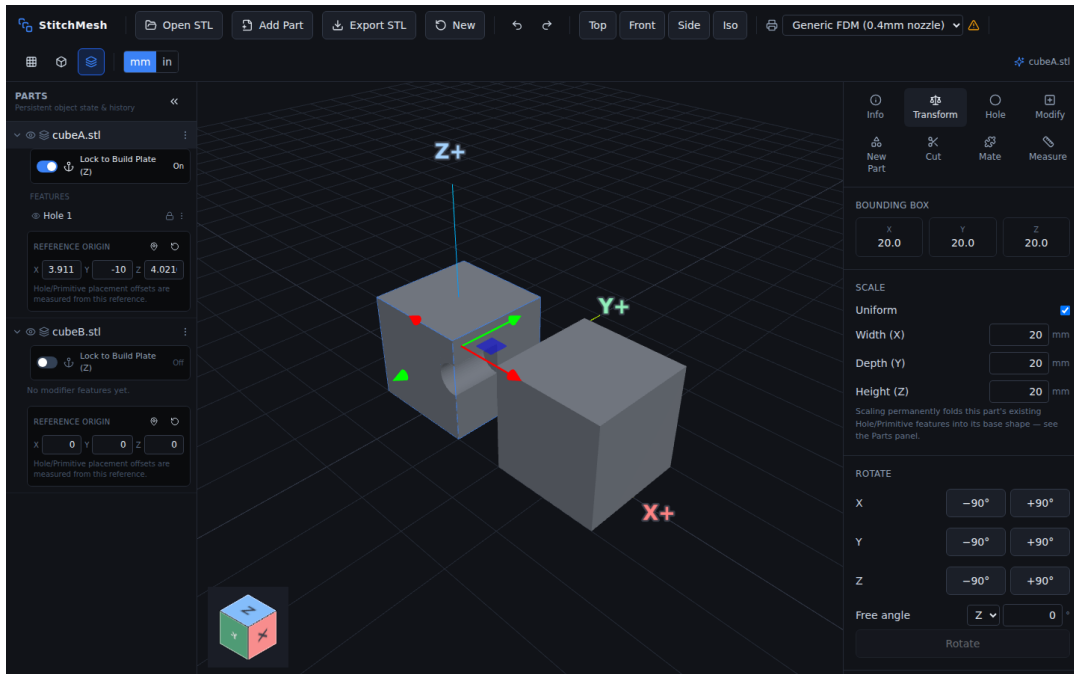
The origin was picked at a corner of `cubeA.stl`. Placing a feature then shows **Offset from local origin X/Y/Z fields** — type an exact position instead of relying on where you clicked.

This is the tool for the "block with two holes a precise quarter-inch from each edge" case: set the origin at a bottom corner of the face you care about, click roughly the right face to establish the drilling direction, then type the exact X/Y/Z offsets for each hole.

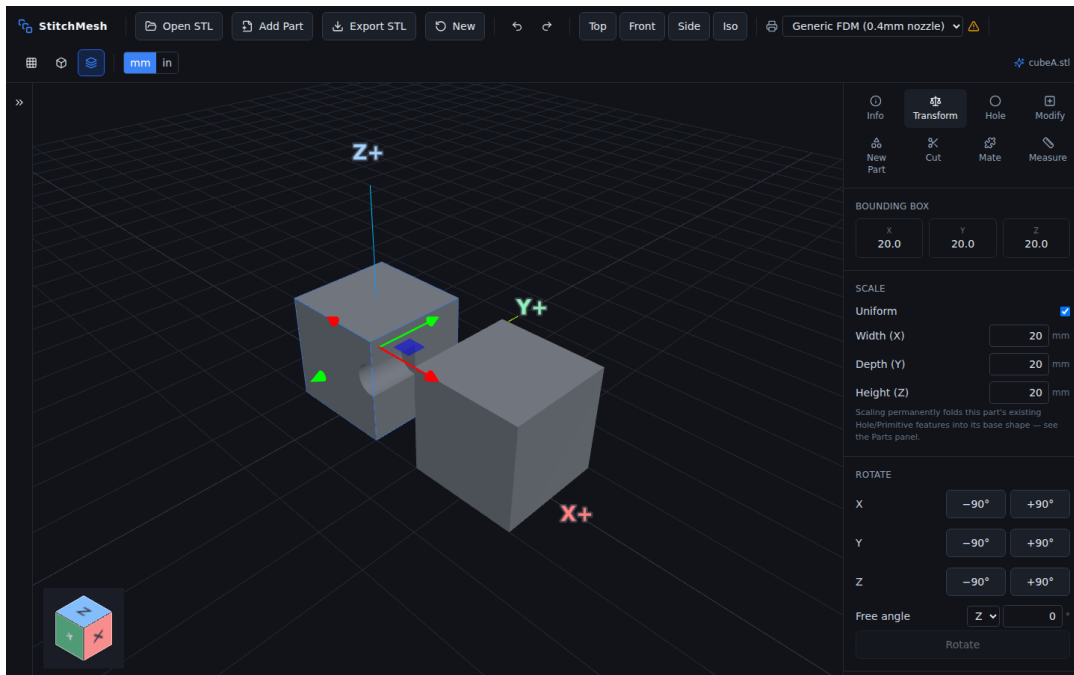
Lock to plate

The **Lock to Build Plate (Z)** switch directly beneath each part name toggles **Lock to plate**. Turning it on immediately drops that part to the build plate (Z=0) and keeps it pinned there afterward:

- The Move tool's Z field and Z-nudge buttons become disabled for that part, and its viewport drag gizmo hides its Z handle.
- You can **left-click-drag the part directly in the viewport** — no gizmo arrows needed — and it slides across the plate in X/Y only. See §12 (Move tool).



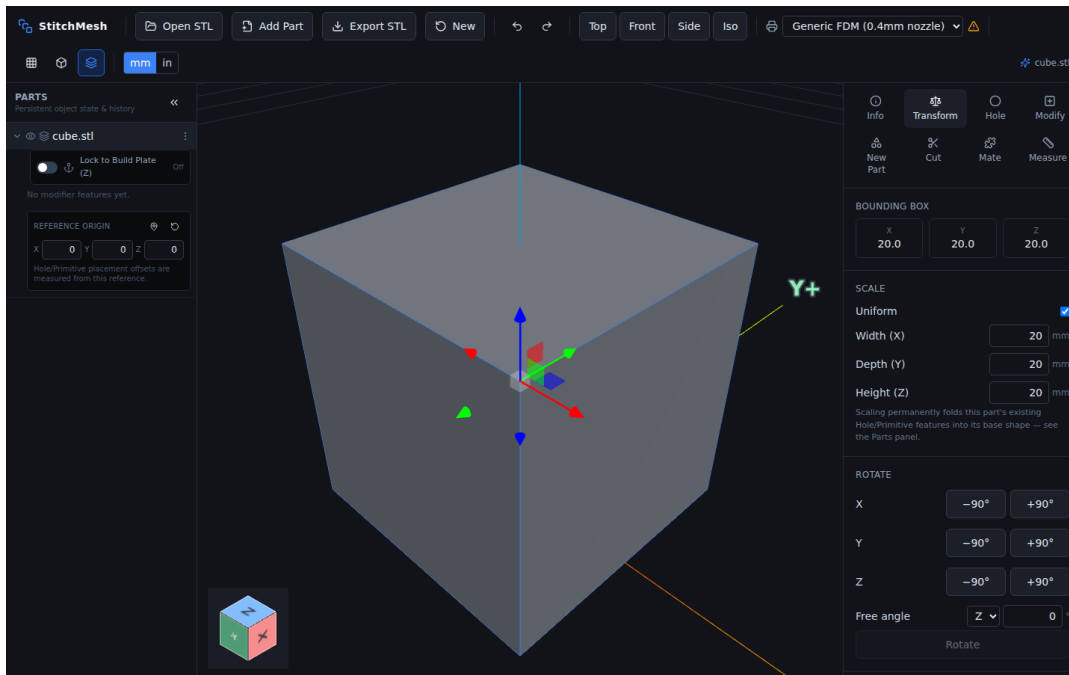
`cubeA.stl` is locked to plate: the Move panel's Z field is disabled with an explanatory note, and the drag gizmo on the model shows only the red (X) and green (Y) arrows — no blue Z arrow.



Collapsed to a thin strip via the « button — click the arrow to expand it again. Purely a display toggle; nothing about the scene changes.

7. Transform tool

Tab: **Transform**. Affects only the **selected part** — every other part in the scene is left alone.



- **Scale** — X/Y/Z fields show the selected part's current size; typing a new value rescales it to that exact size. **Uniform** (on by default) keeps all three axes proportional when editing one field. See §6 for how scaling interacts with that part's existing features.
- **Rotate** — $\pm 90^\circ$ snap buttons per axis, plus a **free-angle** row: pick an axis, type any number of degrees, click **Rotate**. Rotating re-drops the part to the build plate afterward.
- **Center to Origin** — recenters that part's bounding box at world (0,0,0) on all three axes.
- **Drop to Build Plate** — Z-only: shifts so the part's lowest point sits at Z=0. Runs automatically once on import too.
- **Mirror** — flips the part across X, Y, or Z through its own center, with corrected winding, so it flips exactly in place and raycasting/ further edits keep working correctly afterward.

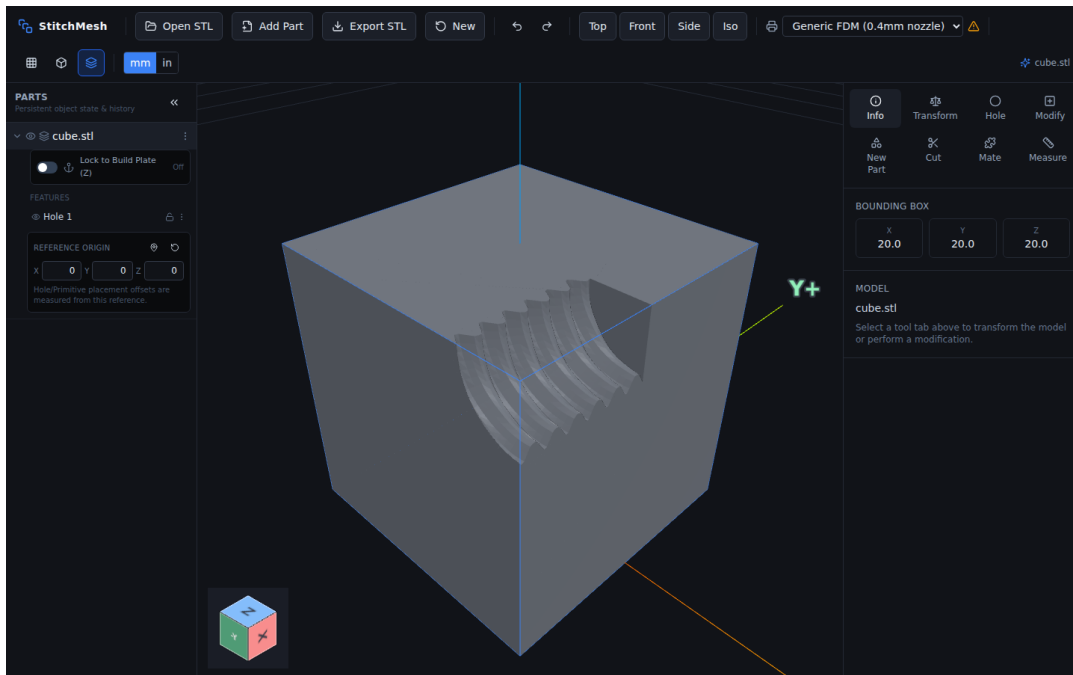
8. Hole Modifier

Tab: **Hole**. Cuts a cylindrical (or threaded) hole at a point you click.

Workflow: click **Hole** → click a point on any part's surface → the sidebar shows Diameter, Depth, Thread, an offset/center readout, and **Apply Boolean Subtract / Cancel**. The part you clicked becomes the target automatically — no extra selection step, even with multiple objects in the scene. The result is saved as its own editable layer under that part — see §6 to reopen, lock, or delete it later.

- **Diameter / Depth** — free-typed when Thread is "Smooth (no thread)". **Depth means depth:** the full stated value is removed from the material, drilling *into* the surface from the clicked point (not split half in/half out).
- **Thread** — a dropdown of standard hardware sizes (§17). Picking one locks Diameter to that thread's exact major diameter and cuts a real helical thread instead of a smooth cylinder.

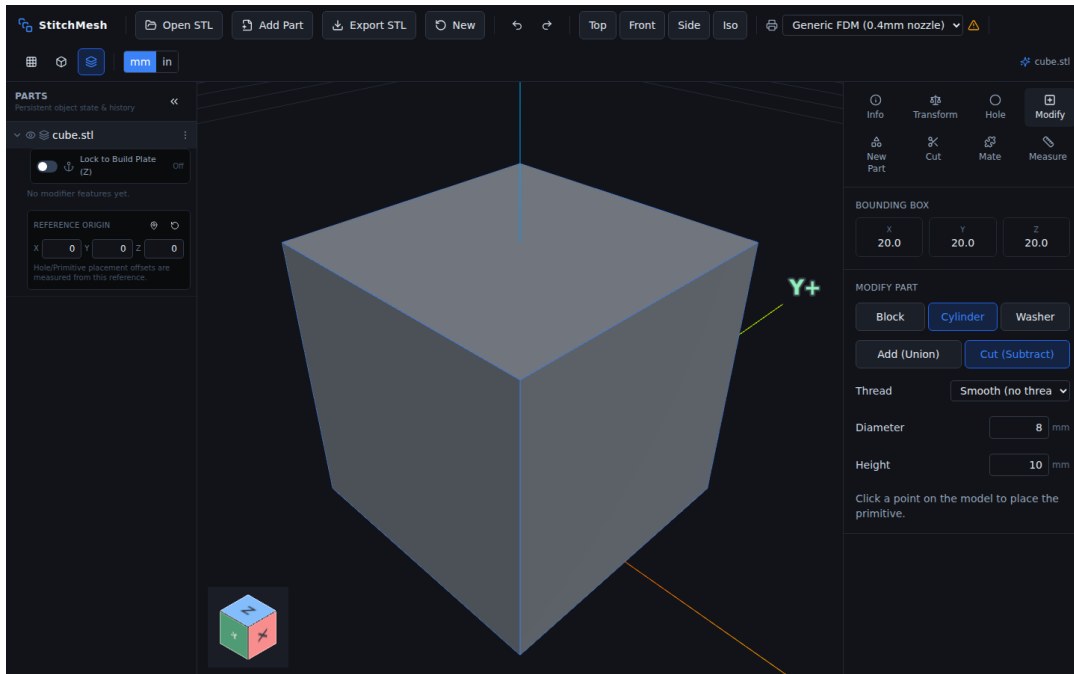
- Δ **from part center / Offset from local origin** — two reference readouts (§6) once the cutter is placed. The offset fields are editable: type exact X/Y/Z numbers to nudge the hole precisely instead of re-clicking.



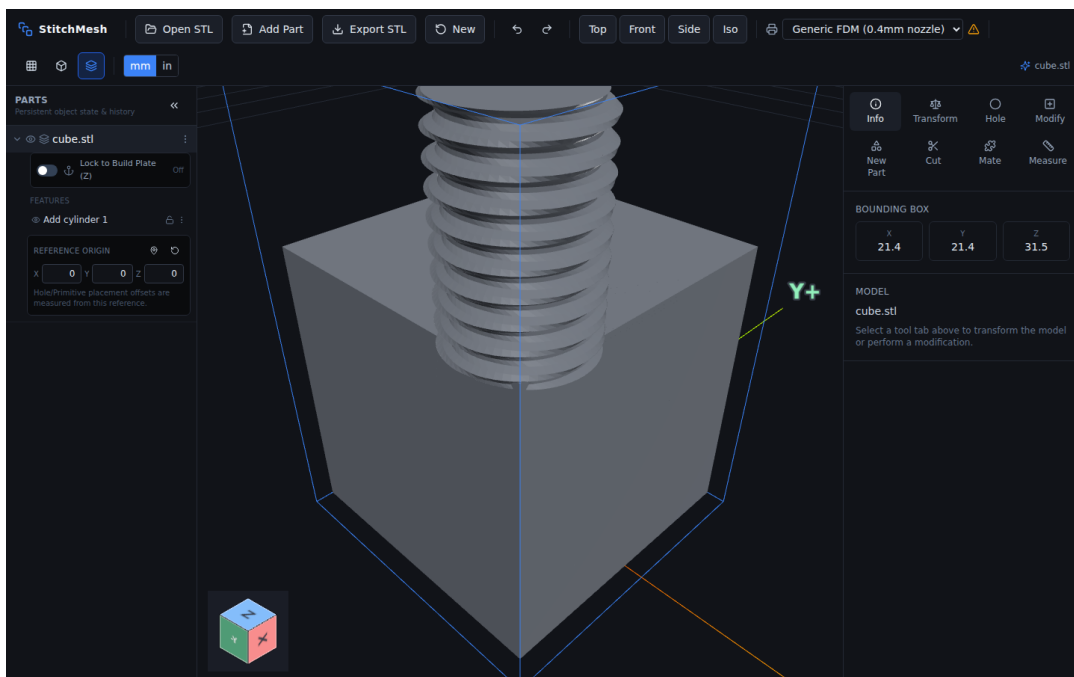
3/8-16 UNC hole applied: a visible internal thread spiraling down the bore, generated from the same geometry a real tap would cut.

9. Primitive Add/Subtract

Tab: **Primitive**. Places a block, cylinder, or washer and unions (adds) or subtracts it from a part at a clicked point — same click-to-place, auto-targeting flow as Hole, and it's saved the same way: as its own editable, lockable, deletable layer under the target part (§6).



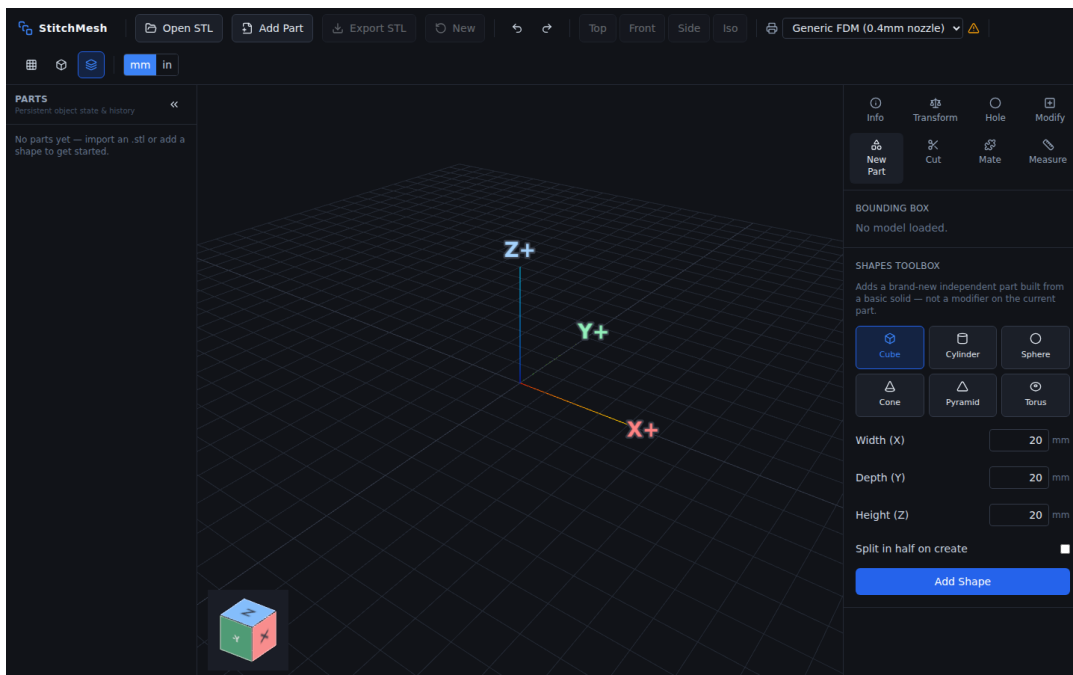
- **Shape:** Block / Cylinder / Washer.
- **Operation:** **Add (Union)** fuses it onto the model as a new protrusion; **Cut (Subtract)** carves it out as a cavity.
- **Dimensions:** Block gets Width/Depth/Height; Cylinder gets Diameter/Height; Washer gets Outer/Inner Diameter/Height.
- **Thread** (Cylinder only): same dropdown as Hole. With **Add** it becomes an external thread (a boss/stud); with **Cut** an internal thread (a tapped hole) — the caption updates to say which.



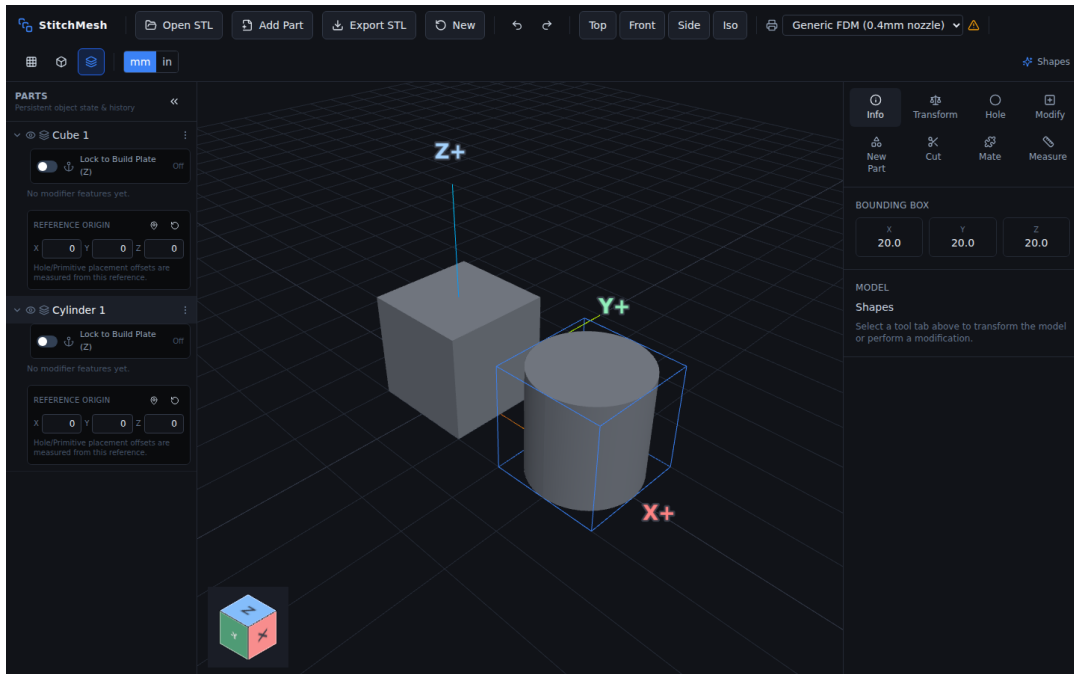
M8 × 1.25, Add (Union), 12mm tall, placed on the top face. It shows up in the Parts panel as "Add cylinder 1" under the target part.

10. Shapes toolbox

Tab: **Shapes**. Adds a brand-new **independent part** built from a basic solid — different from Primitive (§9), which modifies an existing part.



- **Shape:** Cube, Cylinder, Sphere, Cone, Pyramid, or Torus.
- **Dimensions:** fields change to match the shape (Width/Depth/Height for a cube; Diameter for a sphere; Diameter + Height for a cylinder, cone, or pyramid; Outer + Tube Diameter for a torus).
- **Split in half on create** — an optional checkbox with an axis picker: creates the shape and immediately slices it into two separate parts along that axis (the same math as Plane Cut, §13), sitting exactly where the whole shape would have been.
- **Add to Scene** (or **Add Shape** if nothing's loaded yet) — drops the new part beside whatever's already in the scene, or centered and on the build plate if it's the first thing you've added.



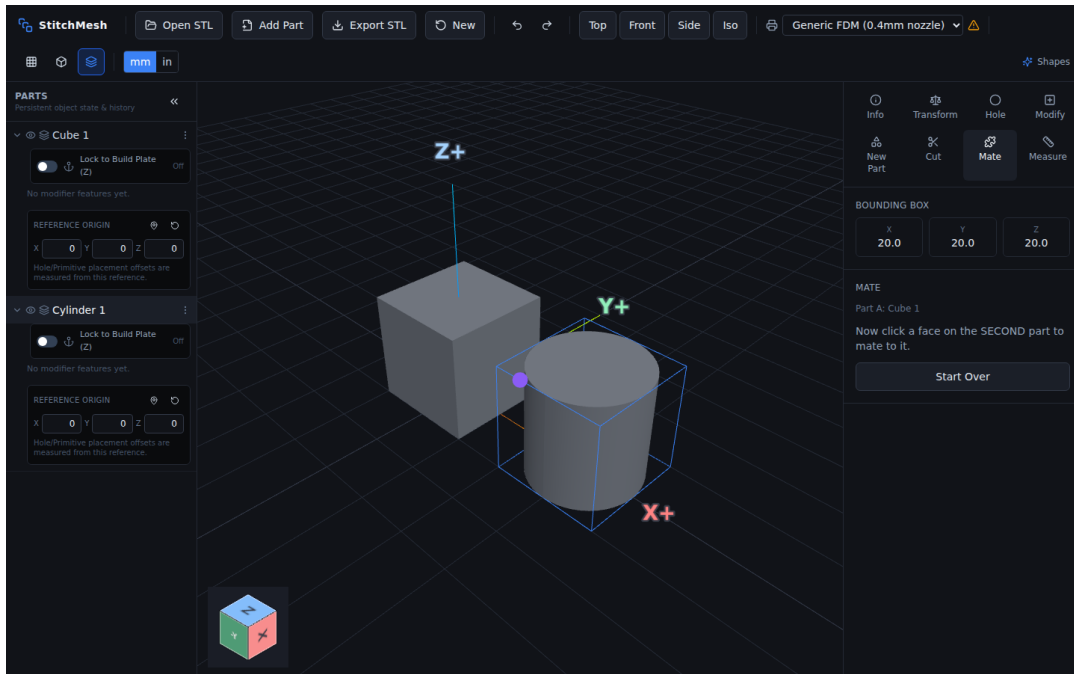
A Cube and a Cylinder, each its own independent part in the Parts panel — ready to be moved, mated, or modified separately.

11. Mate & Weld

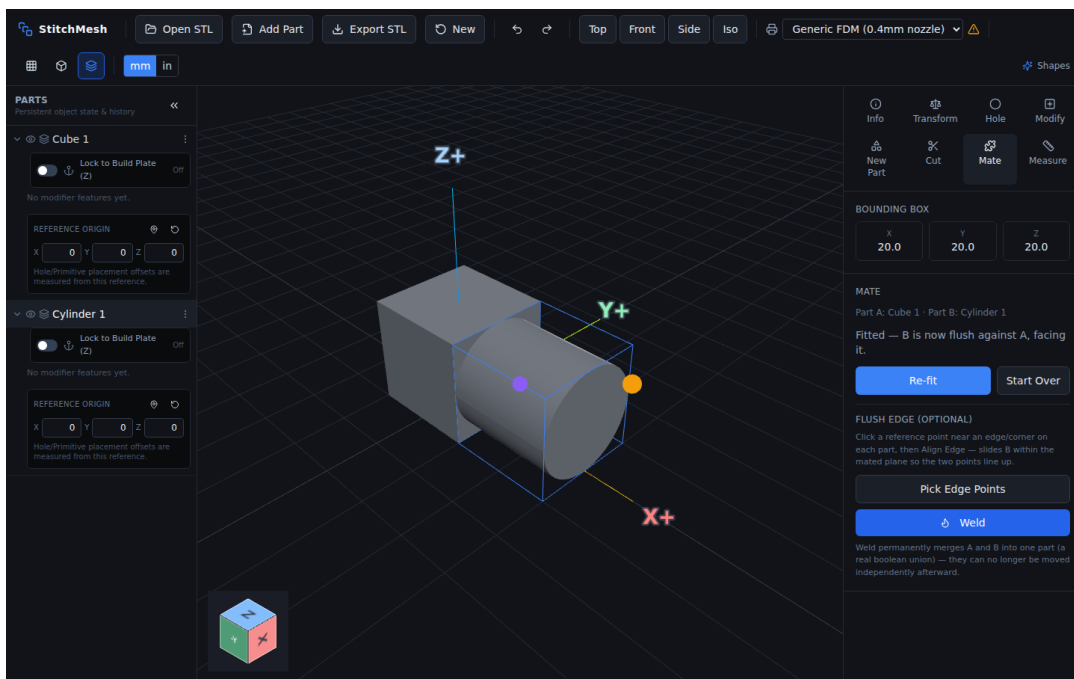
Tab: **Mate**. "Smart-fits" two parts flush against each other — a face on one part is rotated and slid until it touches a face on the other, facing it — while keeping them as two separate, independently movable parts. **Weld** then permanently merges them, if you want that.

Workflow:

1. Click **Mate** → click a face on the **first (stationary)** part.
2. Click a face on the **second** part — the one that will move.
3. Click **Fit** — StitchMesh rotates and slides the second part so that face sits flush against the first, facing it.



After clicking a face on the first part, click a face on a different part to mate it to.



A cylinder fitted flush against a cube's face — still two separate parts in the Parts panel. **Re-fit** repeats the fit if you've moved something; **Start Over** clears the picks and starts again.

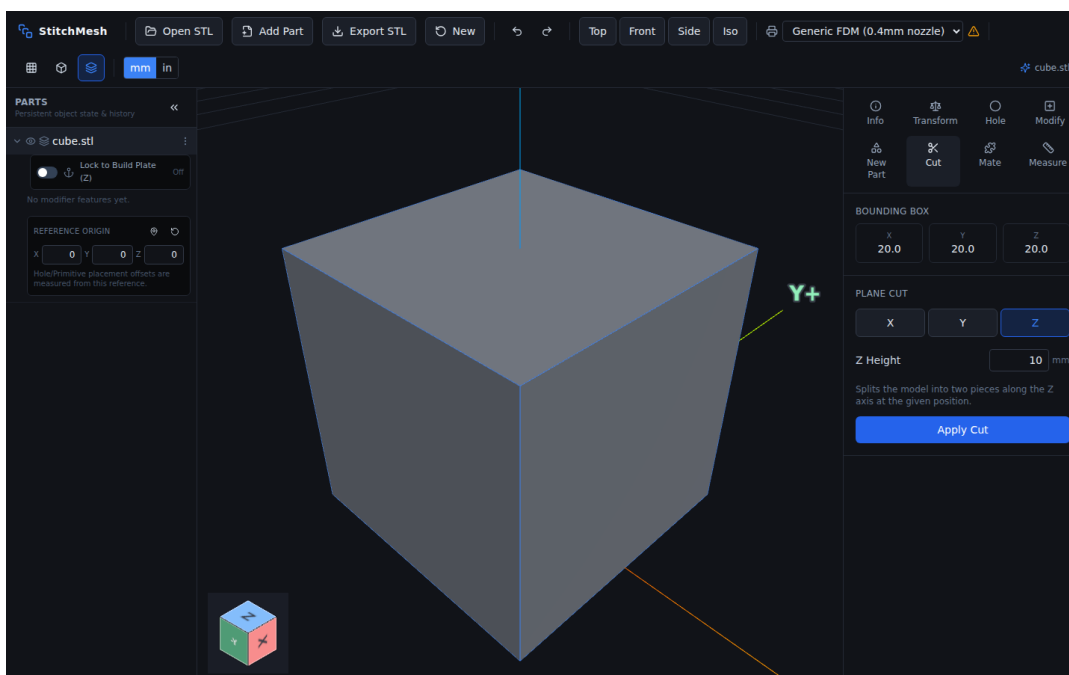
- **Flush Edge (optional)** — once fitted, click **Pick Edge Points**, then click a reference point near an edge or corner on each part. **Align Edge** then slides the second part *within* the mated plane (the flush contact from Fit is preserved) so those two points line up — useful for centering a boss on a face or aligning a corner exactly. This is a point-to-point alignment, not true edge detection, so pick points as close to the actual edge/corner you mean as you can.

- **Weld** — a real boolean union that permanently merges the two mated parts into one. They can no longer be moved independently afterward, and neither part's prior feature layers (§6) carry over as separately editable on the merged result.

Because StitchMesh works on triangle meshes rather than true CAD faces, a "face" here means whatever flat surface you click, and "flush" means the two clicked surfaces' planes touch with their normals pointing at each other — it works well for flat faces on blocks, bosses, and similar shapes, and won't recognize curved or compound surfaces as a single face the way a parametric CAD face would.

12. Plane Cut

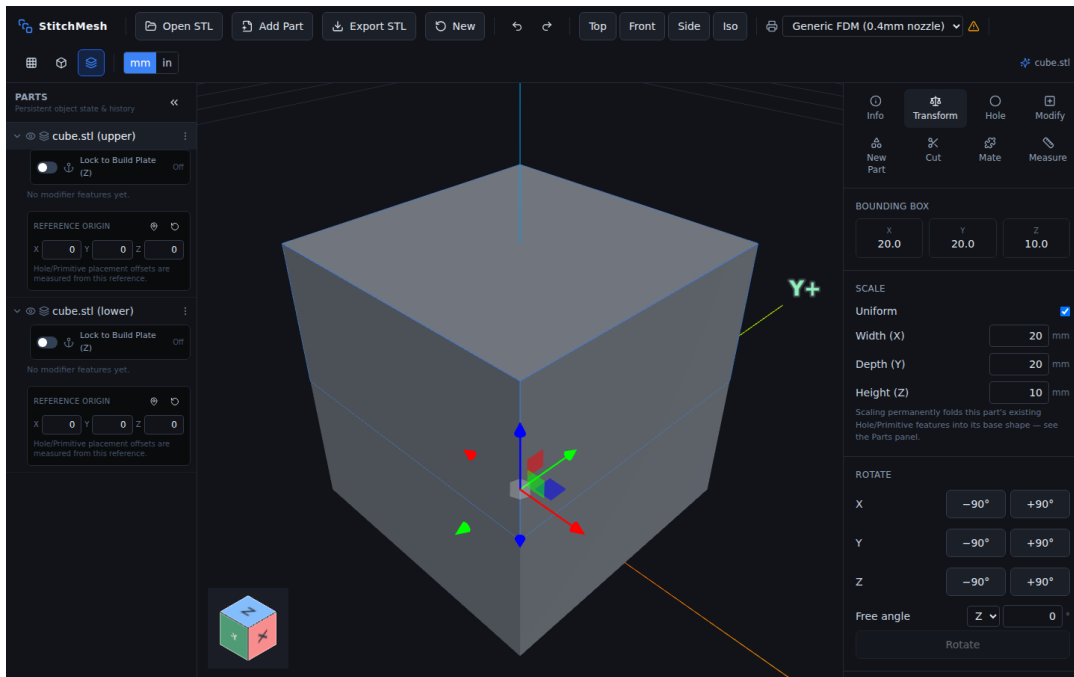
Tab: **Cut**. Splits the target part into **two independent, separately selectable parts** along an axis-aligned plane.



- **Part selector** — only shown once more than one part exists.
- **Axis** — X / Y / Z, the cutting plane's normal.
- **Height** — the plane's position on that axis. Auto-defaults to the target's actual center whenever you open the tool or change axis (a model sits on the build plate, so a fixed 0 default would cut at the very bottom edge and remove nothing).
- **Apply Cut** — both resulting pieces are kept, appear as two new entries in the Parts panel labeled (**upper**) and (**lower**), and are exported together. They start out sitting exactly where the original was — see the Move tool to actually pull them apart. Neither piece carries over the original part's feature layers as editable — the cut bakes the shape as it stood at that moment into each new piece's own base.

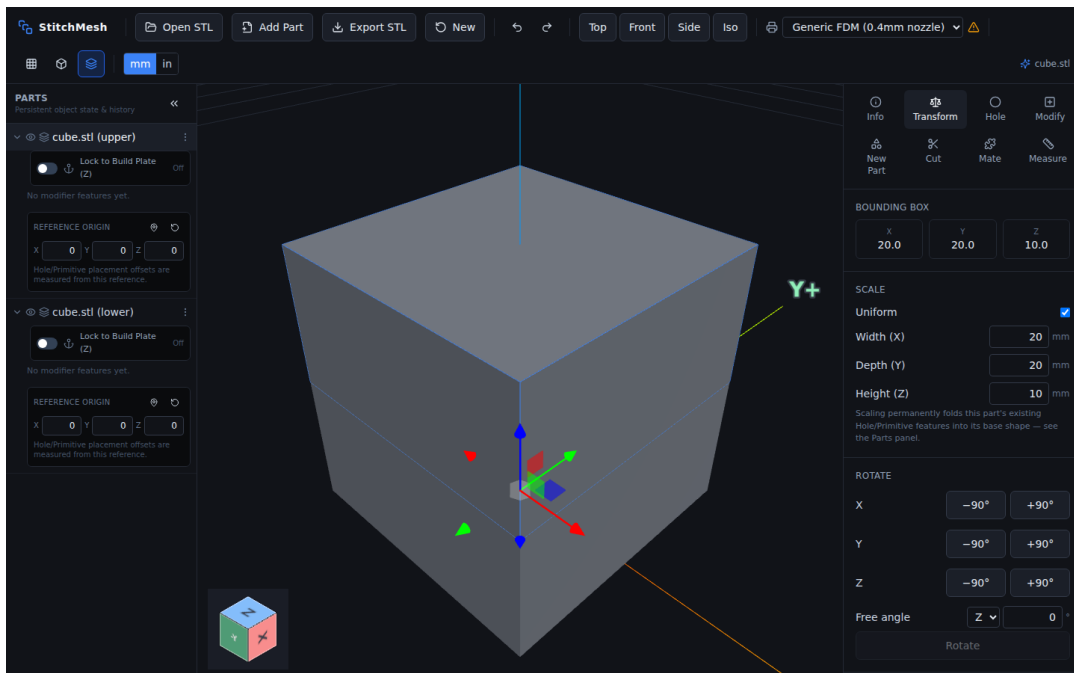
13. Transform and Move

Tab: **Move**. Selects a part and repositions it — either directly in the viewport, or with the numeric fields.



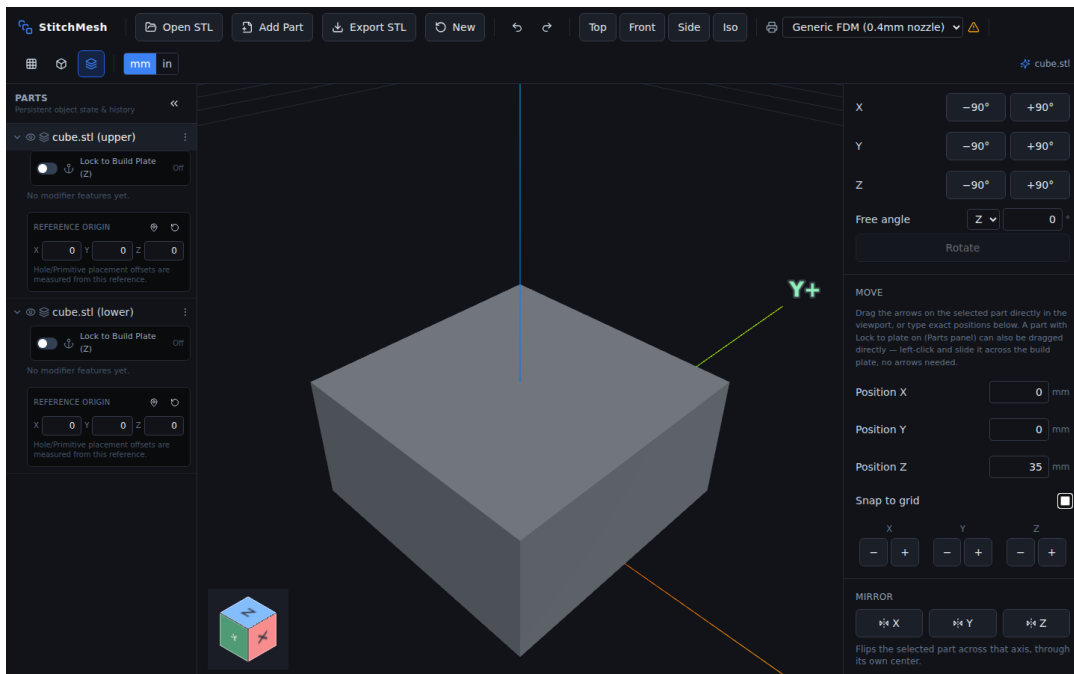
Right after a Plane Cut: two selectable parts exist in the Parts panel (`cube.stl (upper)` / `(lower)`), but they still occupy the same space. Their positions can be separated directly from Transform.

- **Drag gizmo** — the selected part gets colored arrows (and a small plane handle) directly in the viewport. Drag an arrow to move along just that axis, or the plane handle to move in two axes at once.
- **Left-click-drag directly on a locked part** — if the part has **Lock to plate** on (§6), you don't need the gizmo arrows at all: just left-click anywhere on the part and drag — it slides across the build plate in X/Y, Z always pinned.



Red (X) and green (Y) arrows and a blue plane-handle square on the selected part. Since this part is locked to plate, there's no blue Z arrow — dragging is constrained to the build plate automatically.

- The selected part is taken directly from the Parts panel. There is no separate Move tab to revisit just to change position.
- **Position X / Y / Z** — absolute position; typing a value moves the part there directly. Z is disabled while Lock to plate is on.
- **Snap to grid** — checkbox; when on, typed positions and drags round to the nearest 1mm (or your chosen grid size).
- **Nudge buttons** (–/+ per axis) — step by the grid size.

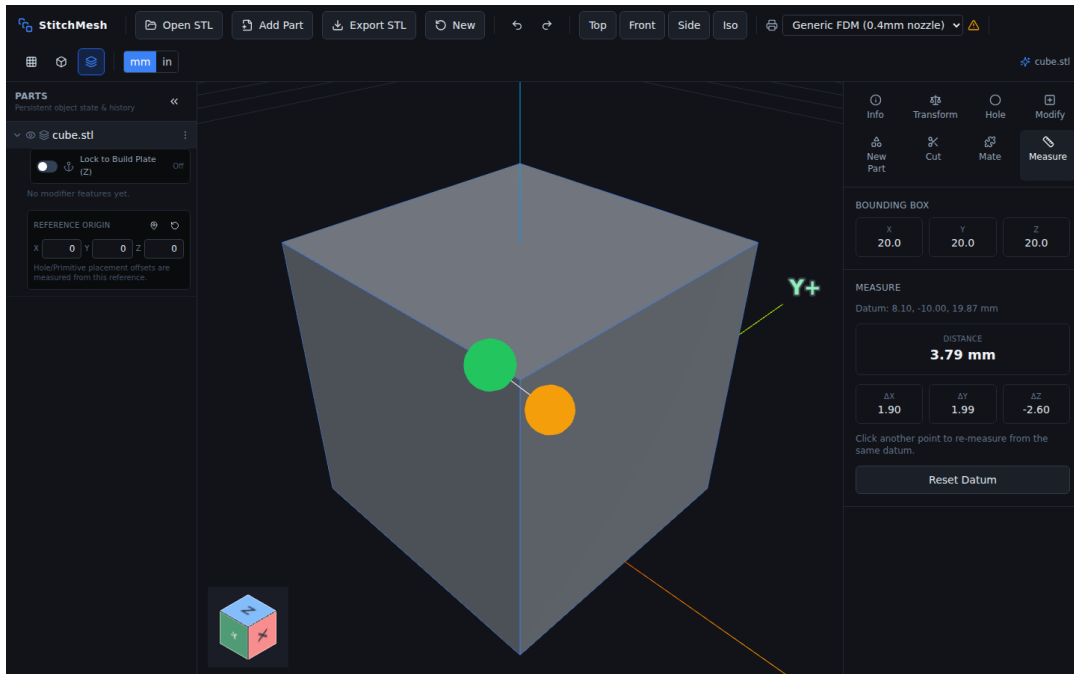


The (upper) piece moved to Z=35: the two halves are now visibly separate, independently selectable, and still export together.

14. Measurement tool

Tab: **Measure**. Click a **datum** (reference point), then click a second point to read the distance and per-axis offset — useful for checking clearances before committing to a hole or thread size.

Workflow: click **Measure** → click a point on the model (sets the green datum marker) → click another point (orange marker + a white connecting line) → the panel shows straight-line **Distance** and ΔX / ΔY / ΔZ . Click again anywhere to re-measure from the same datum without resetting it. **Reset Datum** clears both points.

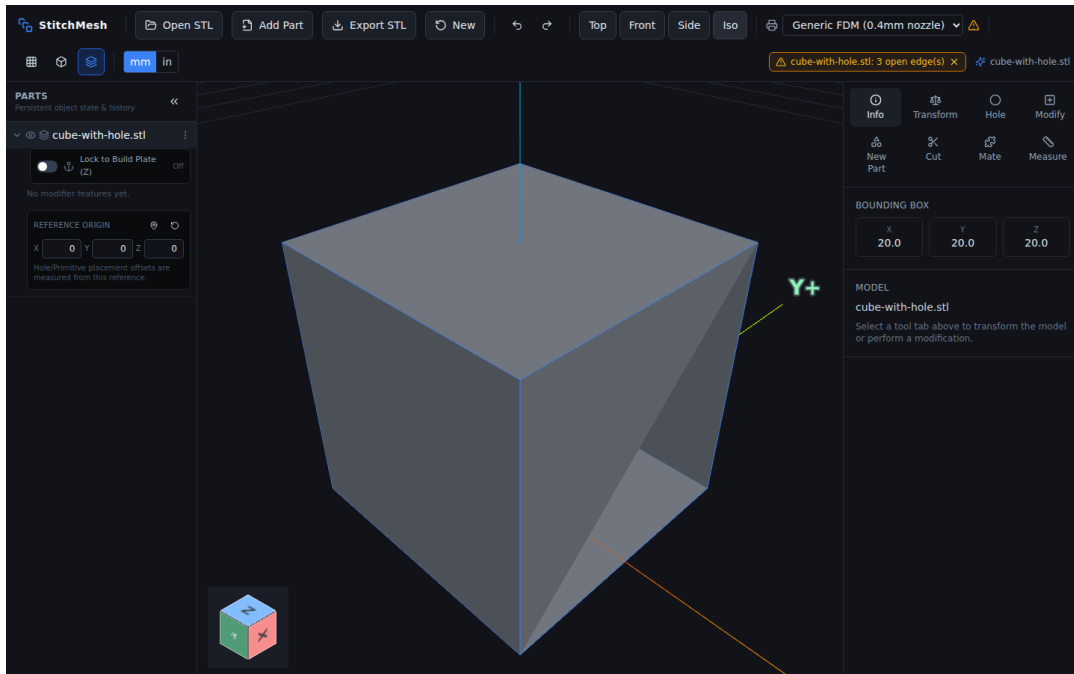


Datum (green) and measured point (orange) on the same cube, with the distance and per-axis deltas shown in the panel.

15. Mesh validation & auto-repair

Every import — the first model or an added part — is checked for two kinds of defect and handled differently:

- **Inconsistent or inverted face winding** (a common real-world defect — a few faces flipped, or occasionally an entire mesh reading inside-out) is detected and **automatically repaired**, silently, before you ever see the model. This matters because StitchMesh's click-to-place tools read the *actual* face winding to know which way is "outward" — an unrepaired flipped face would drill in the wrong direction. Repair runs in two passes: a local pass that unifies winding within each connected patch of geometry (by majority vote against its neighbors), then a global pass that flips the whole mesh if it still reads inside-out overall.
- **Open boundary edges** (holes) or **non-manifold edges** (edges shared by three or more triangles — not a valid solid) are **not** auto-fixed, since closing a hole means guessing a shape. Instead, a dismissible banner appears in the toolbar.



A model with one triangle deliberately removed: "cube-with-hole.stl: 3 open edge(s)" with a dismiss (X) button. The banner is informational only — StitchMesh doesn't attempt to fill the hole automatically.

16. Thread generator & standard sizes

Both the Hole and Primitive (cylinder) tools share one **Thread** dropdown with three groups:

- **Metric (ISO):** M2, M2.5, M3, M4, M5, M6, M8, M10, M12, M14, M16, M18, M20 (coarse pitch)
- **Unified Coarse (UNC):** #4-40, #6-32, #8-32, #10-24, 1/4-20, 5/16-18, 3/8-16, 7/16-14, 1/2-13, 5/8-11, 3/4-10
- **Unified Fine (UNF):** #4-48, #6-40, #8-36, #10-32, 1/4-28, 5/16-24, 3/8-24, 7/16-20, 1/2-20, 5/8-18, 3/4-16

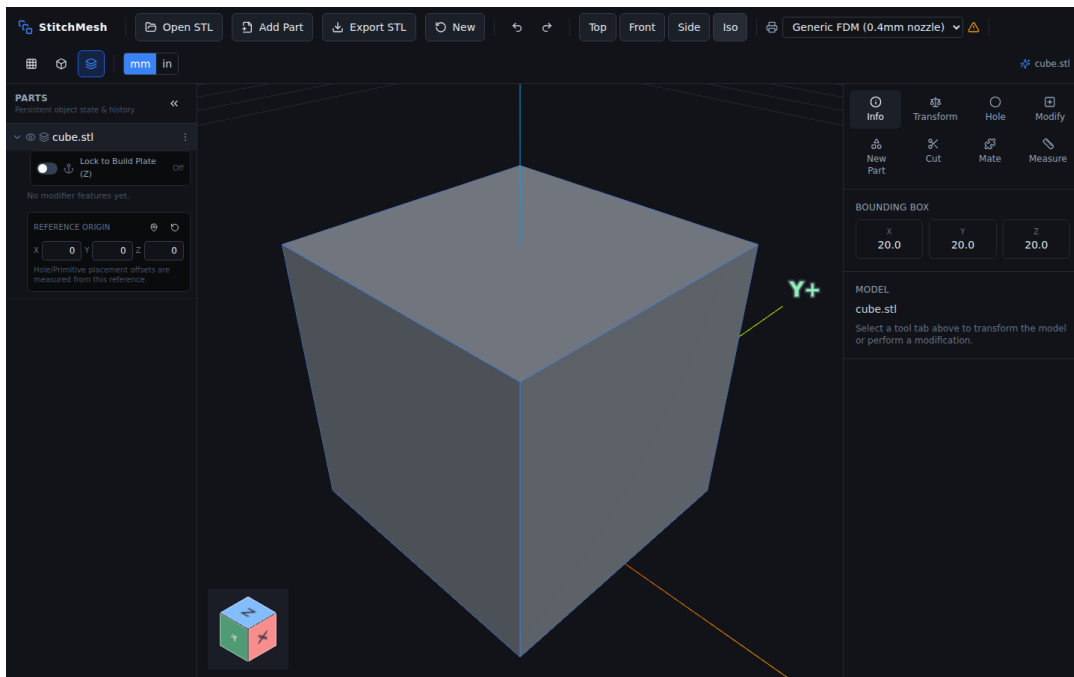
Picking a size generates an actual helical 60° V-thread mesh, from the fundamental-triangle geometry both standards share (ISO 68-1 for metric, ASME B1.1 for Unified):

```
H (fundamental triangle height) = 0.866025 × pitch
pitch diameter                  = major diameter - 0.649519 × pitch
minor diameter                  = major diameter - 1.082532 × pitch
crest truncated by H/8, root truncated by H/4
```

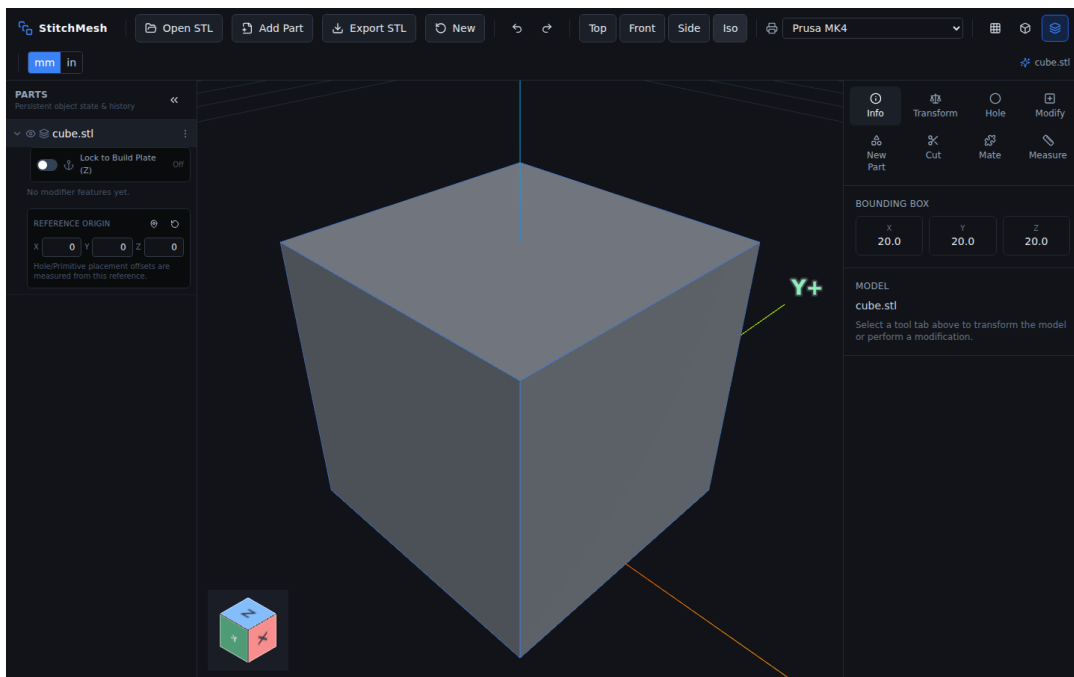
These constants were cross-checked against known published table values (M6×1.0 → minor diameter 4.917mm; M10×1.5 → pitch diameter 9.026mm) — both matched exactly. An internal (tapped) thread and an external (screw) thread are the *same* nominal profile — the generator is reused as a subtraction cutter for one and a union addition for the other, exactly like a real tap cuts the mating shape of the bolt it's sized for. See [src/utils/threadGeometry.ts](#) for the profile math and [src/utils/threadStandards.ts](#) for the size table.

17. Printer profiles

The toolbar's **Printer** dropdown tunes every threaded feature to a specific machine:



Default: "Generic FDM (0.4mm nozzle)" with a warning triangle — shown whenever Generic is active, confirmed or not, since the underlying concern (no machine-specific tuning) is still true either way.



After picking "Prusa MK4": the warning icon clears.

What the profile changes:

- **Mesh resolution** — radial facets and helical height-rings sized to the printer's nozzle/spot diameter and typical layer height.
- **Internal-thread clearance** — a small diametral clearance added to tapped holes (0.25–0.35mm for FDM depending on machine, 0.1mm for resin) so a print accepts a real bolt despite typical over-extrusion.
- **Printability warning** — in the Thread panel, when a pitch is finer than the nozzle can resolve, or the size is small enough (below roughly M4/#8) that a threaded insert would print more reliably.

Included: Generic FDM, Bambu Lab X1 Carbon, Bambu Lab P1S, Prusa MK4, Creality Ender 3 V2, Creality K1C, Voron 2.4 (350mm), Ultimaker S5, Elegoo Saturn 3 (resin), Formlabs Form 4 (resin).

18. Why the Parts-first workflow matters

StitchMesh deliberately separates **persistent object state** from **temporary operations**:

- The **Parts panel** answers: *What objects are in the scene, which are visible, which are protected, and what modifications belong to each object?*
- The **viewport** answers: *What am I looking at and where am I clicking?*
- The **right sidebar** answers: *How should the current operation behave?*

This keeps the interface from becoming a second toolbox inside the object list. Persistent state stays visible on the left; operation parameters stay on the right. Context menus provide shortcuts without duplicating every tool in every part row.

Recommended workflow

1. Select a part in the Parts panel.
2. Use the eye/anchor/lock controls to establish its persistent state.
3. Right-click the part for a direct shortcut such as **Add hole**, **Modify part**, or **Transform**.
4. Configure the operation in the right sidebar.
5. Name important features so a later edit is understandable.
6. Hide or isolate other parts when inspecting a tight assembly.
7. Use **Measure** before committing a dimension that must fit another part.

19. Every control, at a glance

Location	Control	Does
Toolbar	Open STL	Import an .stl , replacing the scene (drag-and-drop works too)
Toolbar	Add Part	Import an .stl as a new independent object
Toolbar	Export STL	Save the current scene as a new .stl

Location	Control	Does
Toolbar	New	Clear the model (confirms first)
Toolbar	Undo / Redo	Step back/forward through edit history (Ctrl+Z / Ctrl+Y)
Toolbar	Top / Front / Side / Iso	Jump to a preset camera framing (1–4)
Toolbar	mm / in	Switch every length field between millimeters and inches
Toolbar	Printer dropdown	Select target printer; tunes thread resolution/clearance
Toolbar	Wireframe / Flat shading / Bounding box	Display toggles (W / — / B)
Parts panel	Part row	Select the active part; expand/collapse its feature history
Parts panel	Eye icon	Show/hide a part
Parts panel	Lock to Build Plate (Z) toggle	Toggle the part's Z-axis lock to the build plate
Parts panel	Part lock icon	Prevent movement/transformation until unlocked
Parts panel	Actions / right-click	Rename, duplicate, isolate, show all, jump to tools, or delete
Parts panel	Feature eye	Show/hide an individual modifier feature
Parts panel	Feature row	Reopen/focus a Hole/Primitive for editing
Parts panel	Feature lock	Freeze/unfreeze a feature against edits
Parts panel	Feature menu	Rename, focus, hide/show, lock/unlock, or delete
Parts panel	Pin / Reset / X,Y,Z	Set, reset, or type a part's reference origin
Parts panel	« / »	Collapse/expand the whole panel
Viewport	Middle-drag	Orbit (re-centers on what's under the cursor)
Viewport	Right-drag	Pan
Viewport	Scroll	Zoom
Viewport	Left-click-drag on a locked part	Slide it across the build plate
Viewport	Bottom-left cube	Shows current view orientation (X/Y/Z labeled faces)
Sidebar → Info	(read-only)	File name, selected part's bounding box
Sidebar → Transform	Scale X/Y/Z, Uniform	Resize the selected part
Sidebar → Transform	Rotate $\pm 90^\circ$ ($\times 3$ axes), Free angle	Rotate the selected part
Sidebar → Transform	Position X/Y/Z, Center, Drop to Plate	Reposition the selected part

Location	Control	Does
Sidebar → Transform	Mirror X/Y/Z	Flip the selected part
Sidebar → Hole	Diameter, Depth, Thread	Configure a hole cutter after clicking a point
Sidebar → Hole	Offset from local origin X/Y/Z	Fine-tune placement numerically
Sidebar → Hole	Apply Boolean Subtract / Save Changes / Cancel / Delete Feature	Commit, update, discard, or remove
Sidebar → Modify	Shape, Operation, dimensions, Thread	Configure a modifier after clicking a point
Sidebar → Primitive	Apply / Save Changes / Cancel / Delete Feature	Commit, update, discard, or remove
Sidebar → New Part	Shape, dimensions, Split in half	Configure a new standalone part
Sidebar → Shapes	Add to Scene	Create the new part
Sidebar → Cut	Part, Axis, Height, Apply Cut	Split the target part in two
Sidebar → Mate	(click two faces)	Pick which parts and faces to mate
Sidebar → Mate	Fit / Re-fit	Rotate + slide part B flush against part A
Sidebar → Mate	Pick Edge Points, Align Edge	Fine-align B within the mated plane
Sidebar → Mate	Weld	Permanently merge A and B into one part
Sidebar → Measure	(click viewport)	Set datum, measure distance/deltas

20. Accuracy & validity review

A full-codebase review (an automated correctness pass, plus manual re-derivation of the placement math for every tool) has been run across several rounds of this project's development. All findings below are **already fixed**.

From earlier passes:

1. **Hole/Primitive depth was silently halved** — the cutter was centered on the clicked point instead of driven into the surface. Fixed; confirmed by drilling a hole deeper than the material and verifying it broke through the far face.
2. **Plane Cut's default height (0) was a silent no-op** on any build-plate-dropped model. Fixed to auto-default to the target's actual center.
3. A coordinate-space bug that could silently miss a boolean entirely after a prior scale/rotate. Fixed.
4. **Transform previously acted on the whole scene instead of the selected part** — scaling or dropping one part after moving another moved or resized both. Root-caused to Hole/Primitive booleans baking each part's *world* transform into its geometry; fixed by compositing features entirely in each part's own stable local frame, verified by scaling/rotating/centering/dropping one part of a two-part scene and confirming the other's position stayed byte-for-byte identical. The same fix closed a latent Mirror bug (mirroring an off-center part used to relocate it instead of flipping it in place).

5. Mesh repair, undo/redo, snap-to-grid, free-angle rotation, and the full feature-layer lifecycle (add/edit/lock/delete) were each verified with dedicated synthetic tests — all correct.

From this pass (controls, Mate/Weld, Shapes, units, lock-to-plate):

- **Negative-number input** — typing a leading "-" (or a bare ".") into any numeric field used to be immediately overwritten back to the old value, because the field re-derived its display text from `parseFloat()` on every keystroke. Fixed by only re-syncing a field's displayed text from its committed value while the field is *not* focused; verified by typing "-1" mid-keystroke and confirming the minus sign survives, then completing "-12.5" and confirming it commits correctly.
- **Units conversion** — verified a 20mm cube reads exactly 0.787in after switching units, and that typed values round-trip correctly in both directions.
- **Mate + Weld** — verified end-to-end: picking a face on each of two parts, Fit (confirmed the parts stay separate, same part count), and Weld (confirmed it merges them into exactly one part).
- **Shapes toolbox** — verified each shape adds a correctly-labeled, independent part, and that "split in half on create" produces two separate (`upper`)/(`lower`) pieces.
- **Lock to plate + left-click-drag** — verified that enabling it snaps the part to Z=0, that left-click-dragging the part (no gizmo arrows needed) moves its X/Y position, and that this same left-click-drag does *nothing* on an unlocked part or empty space (no accidental orbit or move).
- **Orbit control change** — verified left-click-drag no longer rotates the camera at all (pixel-identical before/after), middle-click-drag still orbits, and right-click-drag still pans.
- **Regression pass:** the full previous feature set (parts/layers, mesh repair, Mirror, Plane Cut + Move, undo/redo, Hole feature edit/ lock/delete) was re-run end-to-end after all of the above changes with no failures.

Everything was re-verified end-to-end in a real browser, offline, with **zero console errors** across every test.

Known limitations (not bugs, but worth knowing):

- **No hole-filling.** Mesh repair fixes winding but doesn't attempt to triangulate and close actual holes/non-manifold topology — those are reported, not fixed.
- **Thread mesh triangle counts scale up fast.** A fine-pitch thread over a long depth can generate 10,000+ triangles.
- **Scaling a part flattens its existing features, and Plane Cut, Mirror, and Weld don't preserve feature layers** either — all four operate on the part's current compiled shape and produce a new base with no separately editable feature history.
- **Mate's "Flush Edge" is point-to-point, not true edge detection** — it aligns two clicked reference points within the mated plane, which works well for corners and straight edges on typical hardware shapes but doesn't recognize curved or compound edges as a single entity.
- **No feature reordering.** Features composite in the order you added them; there's no drag-to-reorder yet.
- **Snap-to-grid applies to the Move tool only**, not to Hole/Primitive click placement.
- **Single-level undo/redo stack**, capped at 25 steps.

20. Possible future features

Most of what was on this list in earlier revisions of this manual is now built. What's still genuinely missing, for a future pass:

1. **Hole-filling / non-manifold repair** — closing actual holes, not just fixing winding.
2. **A rotate/scale mode for the viewport gizmo**, alongside the current translate-only drag handles.
3. **True edge/face detection for Mate**, instead of point-and-normal picking.
4. **Feature reordering** — drag a layer up/down in the Parts panel to change the order features composite in.
5. **Text/label embossing** onto a surface.
6. **Per-vertex or sculpting-level editing** — StitchMesh is deliberately scoped to primitive-based modification, not freeform mesh editing.
7. **Saved/named printer profiles** beyond the built-in list (custom nozzle diameter, layer height).

Thanks for using StitchMesh. *Modify, don't model.*